SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMKUR

(A Constituent College of Sri Siddhartha Academy of Higher Education, Agalakote, Tumkur)

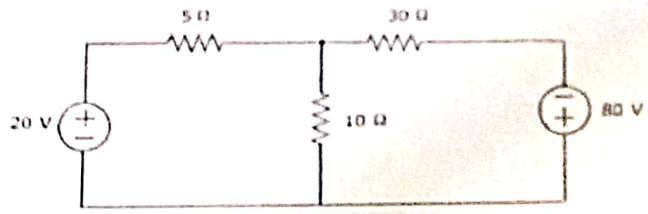
B.E. Test-1 – NOV 2023

22EC302: NETWORK THEORY

TIME: 1.00 Hr **SEM: III** **MAX MARKS: 30**

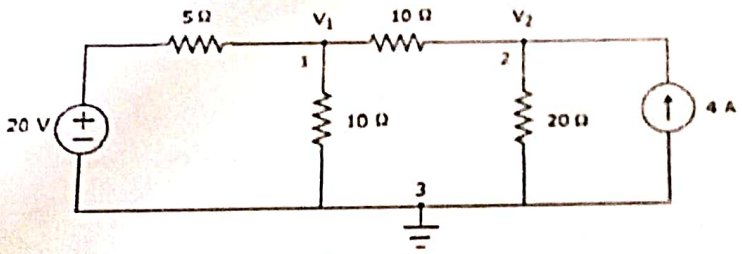
NOTE: Answer all the questions;

		M	CO	BL
1	Define and explain the following terms a) Ideal and practical voltage sources b) Kirchhoff's voltage Law and Kirchhoff's Current law	6	1	2
2	Find the voltage across 10 Ω resistor of the circuit shown in figure (1) using Mesh analysis.	6	1	2



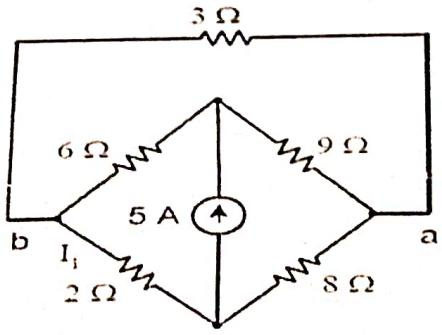
Figure(1)

3	Calculate the current flowing through 20 Ω resistor of the circuit shown in figure (2) using Nodal analysis.	6	1	2



Figure(2)

4	Use source shifting and transformation techniques to find voltage across a; b resistor as shown in Figure (3).	6	1	2



Figure(3)

5	State and explain Thevenin's theorem.	6	2	2

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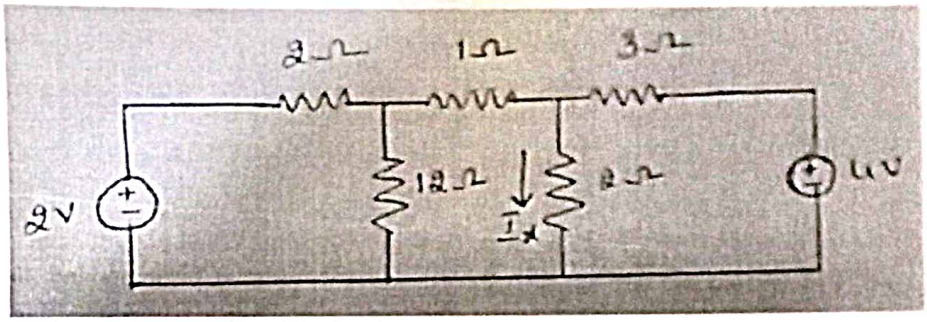
B.E. Test-2 – DEC 2023

22EC302 : NETWORK THEORY

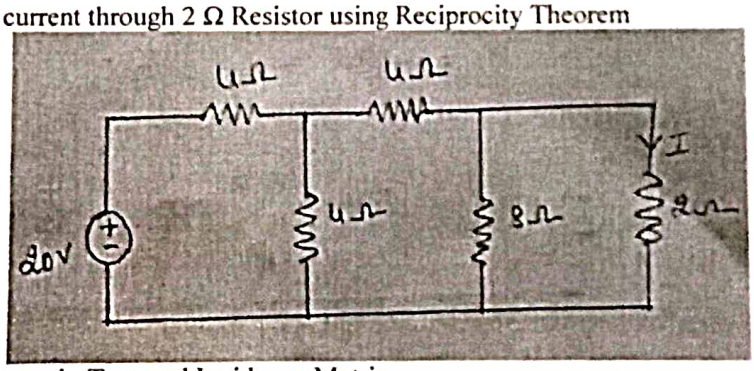
TIME: 1.00 Hr | SEMESTER:III | MAX MARKS: 30

Answer all the questions:

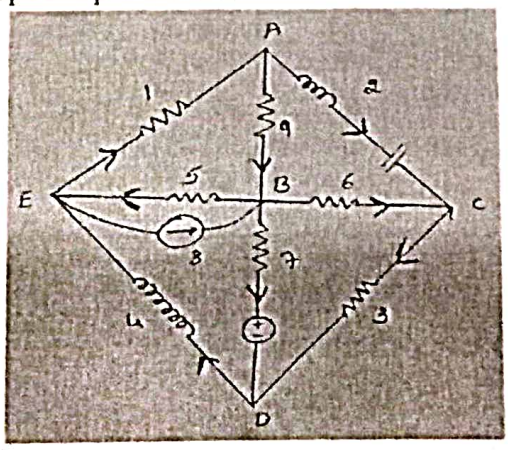
		CO	BL	Marks
1)	Find the current in 2Ω using superposition theorem	2	3	07



2)	State and prove maximum power transfer theorem for DC load, resistive internal impedance and variable resistive load.	2	3	07
3)	Find the current through 2 Ω Resistor using Reciprocity Theorem	2	3	08



4)	a) Define graph, Tree and Incidence Matrix b) Write oriented graph and possible tree	2	1	08



SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMKUR.

(A Constituent College of Sri Siddhartha Academy of Higher Education, Agalakote, Tumkur.)

Department of Electronics and Communication

B.E. TEST-III –JAN-2024

TIME: 60 Minutes **22EC302: NETWORK THEORY**

SEMESTER:III **MAX MARKS: 20**

Answer all the questions:

		CO	BL	Marks
1.	For a series resonance circuit, derive expression to find Resonant frequency and quality factor. In a series resonant circuit R=10Ω, w_0=8.5x106rad/sec and Bandwidth=105rad/sec. Compute L,C and quality factor of the circuit.	2	L2	8
2.	Draw the dual network for a network shown in figure1. Write integro-differential equations for both networks.	3	L3	6

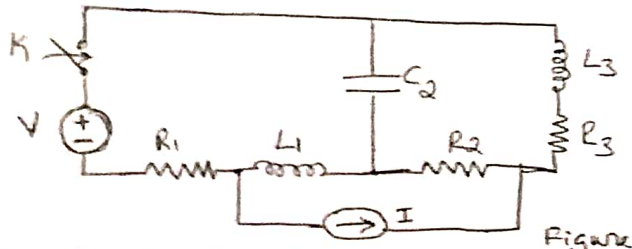


Figure (1)

3.	Explain open circuit impedance (Z) parameter and short circuit admittance (Y) parameter of a two port network.	2	L1	6

2 2 E C 3 2 0

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SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY- TUMAKURU

(A constituent College of Siddhartha Academy of Higher Education, Tumakuru)

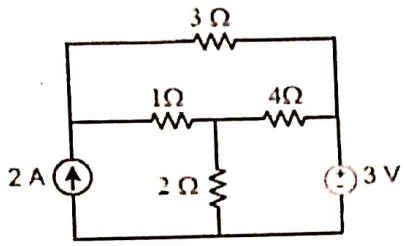
B.E., SEMESTER END EXAMINATION JAN - FEB 2024

22EC302: NETWORK THEORY

TIME: 3.00 Hrs **SEM: III** **MAX MARKS: 100**

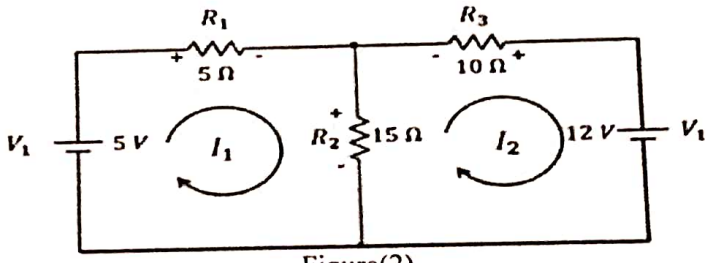
NOTE: Answer any five full questions selecting one full question from each choice.

		M	CO	BL
1.a)	Three resistances are connected in delta, obtain its star equivalent network.	6	1	2
b)	Use source shifting and transformation techniques to find voltage Vx in the circuit as shown in Figure (1).	7	1	2



Figure(1)

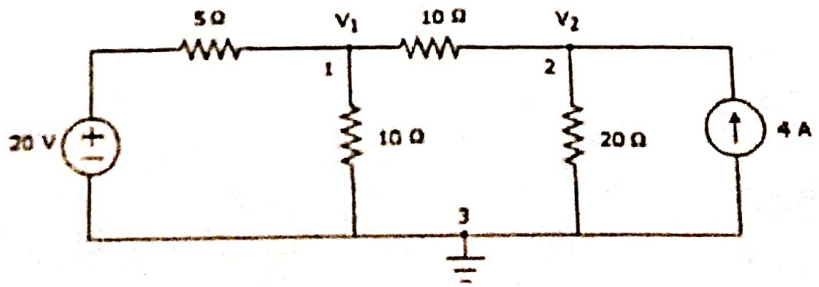
c)	Find the voltage across 10 Ω resistor of the circuit shown in figure (2) using Mesh analysis.	7	1	2



Figure(2)

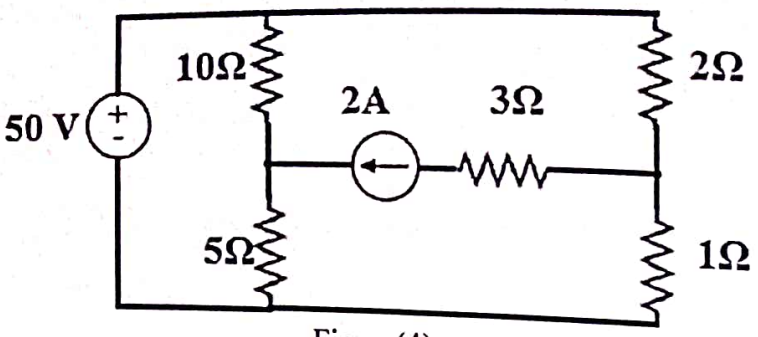
OR

2.a)	Define and explain the following terms a) Ideal and practical voltage sources b) Kirchhoff's voltage Law and Kirchhoff's Current law	6	1	2
b)	Calculate the current through 20 Ω resistor of the circuit shown in figure (3) using Nodal analysis.	7	1	2



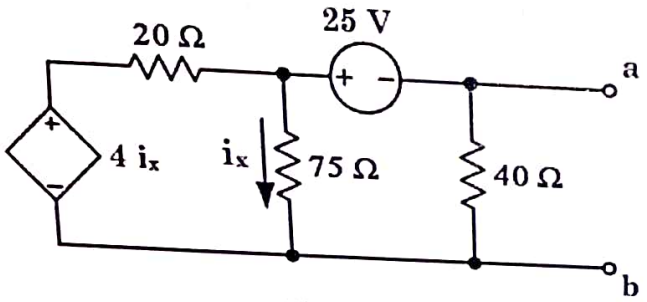
Figure(3)

c) Find the currents through 5ohm resistor in the circuit as shown in figure (4) using Super Mesh analysis. 7 1 2



Figure(4)

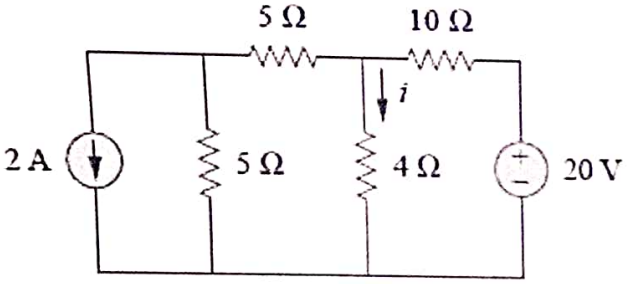
3.a) Obtain Thevenin's equivalent circuit across terminals a,b of the circuit shown in figure (5) 7 2 2



Figure(5)

b) State and explain Millman's theorem 6 2 1

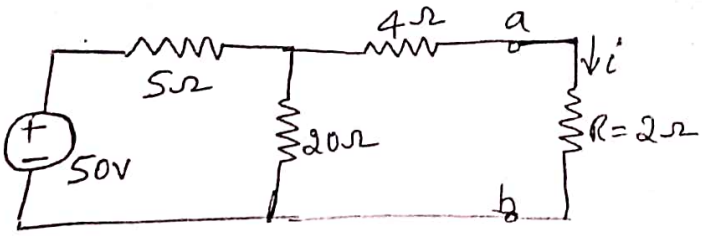
c) Using Superposition theorem, find the current i through R=4Ω, for the circuit shown in figure (6) 7 2 2



Figure(6)

OR

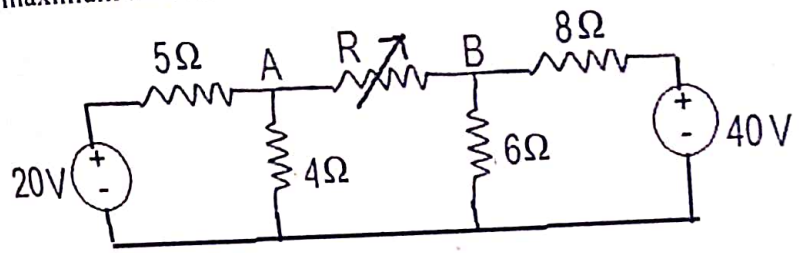
4.a) Find the current i through R=2Ω, for the circuit shown in figure.7. Using Norton's theorem, 7 2 2



Figure(7)

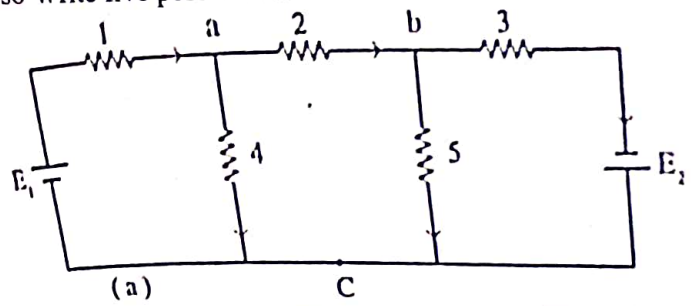
b) State and explain Reciprocity Theorem. 6 2 1

) Find the value of R for which the power transferred across AB of the circuit shown in figure (8) is maximum also calculate the maximum power transferred. 7 2 2



Figure(8)

5.a) Draw an oriented graph and write complete incidence matrix for the network shown in figure (9). Also Write five possible trees. 6 2 2



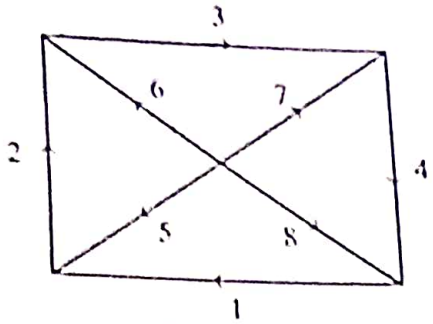
Figure(9)

b) A series RLC circuit has R=20Ω, L=1.5H and C=20μF. The applied voltage is 200V. Find (a) Resonant frequency & Quality factor of a coil (b) Bandwidth (c) Upper and lower Half power frequencies (d) Current at resonance & current at half power points (e) Voltage across inductance & voltage across capacitance at resonance. 10 4 2

c) Explain the concept of cut-set and cut-set schedule. 4 2 1

OR

6.a) For the graph shown in figure (10) write Tie– set matrices (Consider tree branches as 5,6,7,8) and equilibrium equations. 6 2 2



Figure(10)

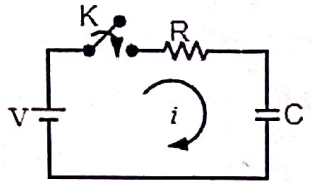
b) Obtain the expression for resonant frequency of a RL-C circuit connected in series. 6 4 2

c) Define and explain the following terms 8 4 1

a) Duality
b) Quality factor
c) Selectivity and bandwidth of series resonance circuit

7.a) State and prove Initial value theorem and Final value theorem. 10 3 2

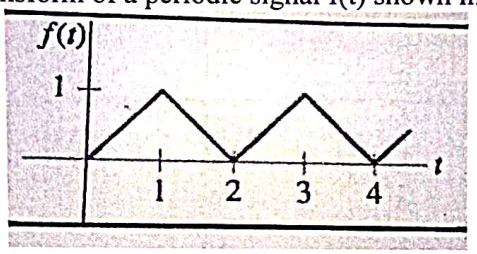
b) In the circuit shown in Figure (11) the switch K is closed at t= 0, with capacitor uncharged. Find the values i, di/dt and d^2i/dt^2 at t= 0+, for element values as follows V=200V, R= 2KΩ and C= 1 μF. 10 3 2



Figure(11)

OR

8.a) Obtain the Laplace transform of a periodic signal f(t) shown in figure (12) 10 3 2



Figure(12)

b) In the circuit shown in Figure (13) the switch K is closed at t= 0, with capacitor uncharged. Find the values of i(0+), di/dt and d^2i/dt^2 for element values as follows, V=10 V R=10 Ω, L= 1 Hand C= 10 μF. 10 3 2

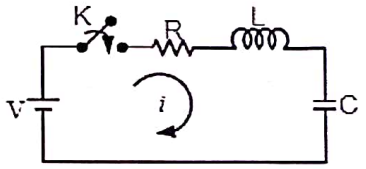
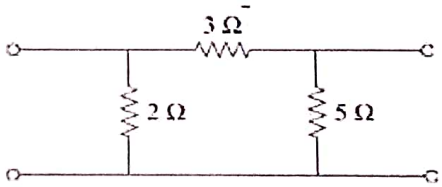


Figure (13)

9.a) Define and Find the Z- parameters of the circuit given in figure(14). 10 4 2

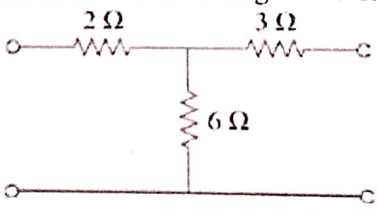


Figure(14)

b) Explain H parameter and ABCD parameter. 10 4 2
Derive H parameters in terms of Z parameters.

OR

10.a) Define and Find the h- parameters for the following circuit shown in figure(15) 10 4 2



Figure(15)

b) Discuss Y parameters, H- parameters and then express H in terms of Y parameters. 10 4 2

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